MATH70082: Probability for Statistics - Coursework

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1 Q1: Semi-algebra

1.1 The case $(a, b]$

- $(0, 1]$ is in this collection, then this collection is not empty. And we should note that $\emptyset$ is in this collection as $\emptyset^c = \Omega = (-\infty, 0] \cup (0, \infty]$.
- We shall satisfy the two properties of this set:
  - Choose $A_1 = (a_1, b_1]$, $A_2 = (a_2, b_2] \cdots A_m = (a_m, b_m]$. Let $a = \max\{a_1, a_2, \cdots a_n\}$, $b = \min\{b_1, b_2, \cdots b_n\}$. If $a < b$, then $\cap_{i=1}^m A_i = (a, b]$. Otherwise, $\cap_{i=1}^m A_i = \emptyset$. However, both cases are all in $\mathcal{F}_s$, then the first property holds.
  - We need to discuss four cases here:
    * If $A = (-\infty, \infty]$, then $A^c = \emptyset = \emptyset \cup \emptyset$.
    * If $A = (-\infty, a]$, then $A^c = (a, \infty] = (a, a+1] \cup (a+1, \infty]$, where $(a, a+1], (a+1, \infty] \in \mathcal{F}_s$ and are disjoint.
    * If $A = (a, \infty]$, then $A^c = (-\infty, a] = (-\infty, a-1] \cup (a-1, a]$, where $(-\infty, a-1], (a-1, a] \in \mathcal{F}_s$ and are disjoint.
    * If $A = (a, b]$, then $A^c = (-\infty, a] \cup (b, \infty]$, where $(-\infty, a], (b, \infty] \in \mathcal{F}_s$ and are disjoint.

    That is to say, for any $A \in \mathcal{F}_s$, A^c could be expressed as the disjoint union of elements in $\mathcal{F}_s$.

Combine these two properties, we say $\mathcal{F}_s$ is a semi-algebra.

1.2 The case (a, b)

Suppose this collection $\mathcal{F}_s$ is a semi-algebra, then for any $A \in \mathcal{F}_s$, A^c is a disjoint union. Choose $A = (-\infty, a)$, then $A^c = [a, \infty)$, which is not open. However, it can be generated by the disjoint union of the open intervals, which leads to a contradiction that the finite disjoint of open intervals is open. Hence, this collection is not a semi-algebra.

1.3 The completion for semi-algebra

The notation $\mathcal{F}_s^c$ represents the completion of $\mathcal{F}_s$.

- (i) As for any $A \in \mathcal{F}_s^c$, A^c is the finite union of sets in $\mathcal{F}_s$, then $\Omega = A \cup A^c \in \mathcal{F}_s^c$.
- (ii) For any $A \in \mathcal{F}_s^c$, $A = \cup_{n=1}^m A_i$, where $A_i \in \mathcal{F}_s$. By De' Morgan's Law, $A^c = \cap_{i=1}^m A_i^c$. From the property of the semi-algebra, all of the A_i^c should be the finite union of disjoint sets in this collection. It is possible for us to start only two disjoint sets, define $A_1^c = \cup_{j=1}^m a_{1,j}$, $A_2^c = \cup_{k=1}^n a_{2,k}$:

$$A_1^c \cap A_2^c = \underbrace{\bigcup_{k=1}^{n} A_1^c \cap a_{2,k}}_{\text{Ensure the decomposed sets are disjoint}} = \underbrace{\bigcup_{k=1}^{n}\bigcup_{j=1}^{m}(a_{1,j} \cap a_{2,k})}_{\text{Ensure the decomposed sets come from } \mathcal{F}_s} \in \mathcal{F}_s^c, \quad (1)$$

  hence, $A_1^c \cap A_2^c$ could be decomposed as finite disjoint sets and further intersects with A_3^c until A_m^c. Hence $A^c \in \mathcal{F}_s^c$. The complement closed property holds.

- (iii) It is enough to verify $A \cup B \in \mathcal{F}_s^c$, provided that $A, B \in \mathcal{F}_s^c$. From (ii), we have that $A^c, B^c \in \mathcal{F}_s^c$. By De' Morgan's Law, $A \cup B = (A^c \cap B^c)^c$, it is sufficient to show $A^c \cap B^c \in \mathcal{F}_s^c$. By the fundamental property of this collection, we have

$$A^c \cap B^c = \underbrace{\bigcup_{k=1}^{n} B^c \cap a_k}_{\text{Ensure the decomposed sets are disjoint}} = \underbrace{\bigcup_{k=1}^{n}\bigcup_{j=1}^{m}(b_j \cap a_k)}_{\text{Ensure the decomposed sets come from } \mathcal{F}_s} \in \mathcal{F}_s^c, \quad (2)$$

  by decomposing $A = \cup_{k=1}^n a_k$, $B = \cup_{j=1}^m b_j$. Hence, we show that $A \cup B \in \mathcal{F}_s^c$.

Combine the above three properties, we show that the completion of a semi-algebra is an algebra.

2 Q2: Uniform Random Variable

From the definition of probability measure, for any set E which is the countable disjoint union of $\{a_i\}$, we have

$$\mathbb{P}(E) = \mathbb{P}(\bigcup_{i=1}^{\infty}\{a_i\}) = \sum_{i=1}^{\infty}\mathbb{P}(\{a_i\}). \tag{3}$$

For each i, we could construct a sequence of intervals A_n, such that

$$A_n = [a_i, a_i + \frac{1-a_i}{n}), \text{where } \bigcup_{n=1}^{\infty} A_n = \{a_i\}. \tag{4}$$

It is also worth to note that $A_{n+1} \subset A_n$, and combine with the convergence of this sequence, we have $A_n \downarrow \{a_i\}$. By referring property for decreasing events of probability measure,

$$\mathbb{P}(\bigcup_{n=1}^{\infty} A_n) = \lim_{n\to\infty}\mathbb{P}(A_n) = \lim_{n\to\infty}\frac{1-a_i}{n} = 0, \tag{5}$$

then $\mathbb{P}(\{a_i\}) = 0$, which is to say $\mathbb{P}(E) = 0$.

3 Q3: Probability Measure

- For any $A \in \mathcal{F}$, if $\mathbb{P}_1(A) \leq \frac{1}{2}$, then $\mathbb{P}_1(A) = \mathbb{P}_2(A)$ from the assumption. On the other hand, if $\mathbb{P}_1(A) \geq \frac{1}{2}$, then $\mathbb{P}_1(A^c) \leq \frac{1}{2}$. Also, as $A \in \mathcal{F}$, due to the complementary closed property of $\mathcal{F}$, we have that $A^c \in \mathcal{F}$ implying that $\mathbb{P}_1(A^c) = \mathbb{P}_2(A^c)$. Hence, we have

  $$\mathbb{P}_1(A) = 1 - \mathbb{P}_1(A^c) = 1 - \mathbb{P}_2(A^c) = \mathbb{P}_2(A). \tag{6}$$

  We exhaust all the cases of A, then we arrive the proof.

- We define the sample space $\Omega = \{0, 1\}$, $\mathcal{F} = 2^{\Omega}$, and then define

  $$\mathbb{P}_1(A) = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1 \quad \textit{and} \quad \mathbb{P}_2(A) = \delta_1, \tag{7}$$

  where δ_a is the *Dirac measure*. Under this circumstance, the only case that $\mathbb{P}_1(A) < \frac{1}{2}$ is $\mathbb{P}_1(\emptyset) = 0$, which is also satisfied by $\mathbb{P}_2$. However, $\mathbb{P}_1(\{1\}) = \frac{1}{2}$ and $\mathbb{P}_2(\{1\}) = 1$, which provides a valid counterexample.